Amar Al-Zubaidi

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github.com/amarz45 · linkedin.com/in/amar-al

EDUCATION

Computer Science, BSc, Hons September 2024 – April 2029

York University, Lassonde School of Engineering North York, Ontario, Canada

- **Cumulative GPA:** 3.7/4.0 (7.85/9.00 institutional scale), member of Dean's Honour Roll
- **Courses:** Data Structures, OOP, Linear Algebra I & II; in progress: Algorithms, Software Design, Numerical Methods I

PROJECTS

Comicpress – github.com/amarz45/comicpress September 2025 – Present

- Built a cross-platform **C++/Qt** application that converts and compresses digital comics for e-readers, with **100+ monthly downloads**
- Designed a libvips-based **image processing** pipeline using worker processes instead of threads to work around PDFium's lack of thread safety, achieving measured throughput that scales linearly with CPU cores
- Shrinks file sizes by **up to 80%** while improving on-device image quality with colour space conversion, quantization, dithering, and resampling configured automatically for the target device
- Ships signed **Windows/Linux** releases automatically with **GitHub Actions** and **Azure Trusted Signing**

ShutterDex – devpost.com/software/htn-vgd2jp Hack the North 2026 (semi-finalist)

- Built an **AI pipeline** turning photos from a custom capture device with an ESP32 microcontroller into 2D creature sprites with chained **Gemini** models for **vision** and **image generation**
- Developed a **custom UI** layer on top of the Hack the North badge's low-level API, enabling creature browsing and battle interactions on constrained **embedded hardware**
- Implemented a **server-authoritative** battle system in **Python/FastAPI** with Wi-Fi client-server communication

WORK EXPERIENCE

Cybersecurity Analyst (Co-op) May 2026 – August 2026

Sobeys Mississauga, Ontario, Canada

- Built an **AI-assisted workflow** analyzing **3 months** of API-retrieved email data to identify **top 5** malware families and campaign trends, with findings verified against source data and used in **monthly executive reports**
- Developed **REST API** scripts in **PowerShell** for CrowdStrike Falcon to automate vulnerability tracking and asset filtering by OS lifecycle status, eliminating a recurring **5+ hour** manual reporting task
- Built automated network scanning using Nmap that classifies devices by combining port, service/version, and OS information, replacing manual per-host identification and enabling fleet-wide EDR sensor rollout

AI Trainer (Code Reviewer) June 2025 – September 2025

DataAnnotation *Remote*

- Reviewed and rewrote AI-generated code in Python, C++, TypeScript, Java, and Rust, implementing requirements the model had silently skipped or left partially complete
- Debugged unfamiliar open-source codebases of **50+ files**, tracing undocumented code paths to locate root causes
- Authored **unit tests** covering edge cases (off-by-one errors, boundary conditions) and evaluation rubrics defining pass criteria for code-correctness tasks, both used by other reviewers to grade submissions consistently

SKILLS

- **Programming languages:** Python, C, C++, Java, TypeScript/JavaScript, Bash, PowerShell
- **Tools:** Git, GitHub Actions, Linux, Claude Code, Copilot
- **Spoken languages:** English (fluent), French (intermediate)